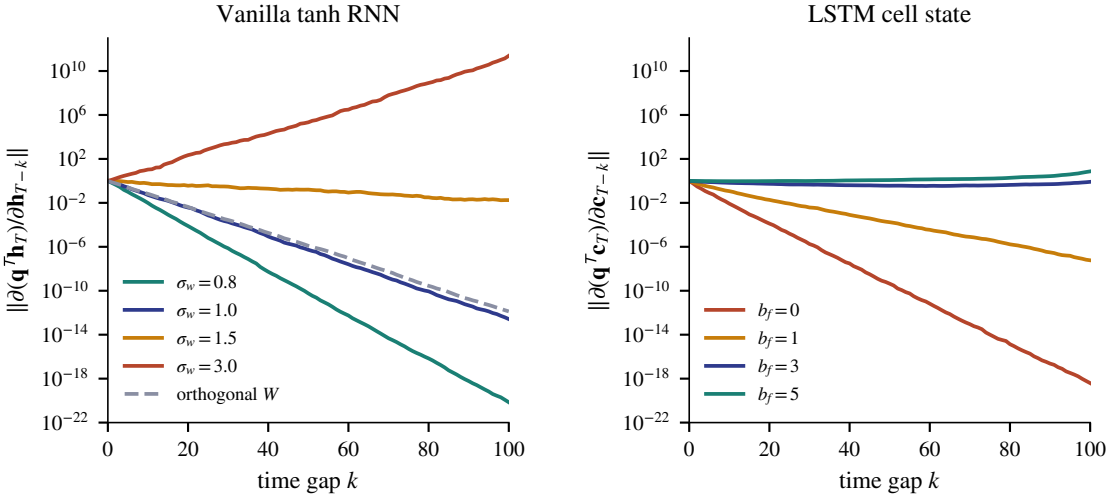
Vanilla tanh RNN
$\|\partial(\mathbf{q}^T\mathbf{h}_T)/\partial\mathbf{h}_{T-k}\|$
10^{10}
10^{6}
10^{2}
10^{-2}
10^{-6}
10^{-10}
10^{-14}
10^{-18}
10^{-22}
$\sigma_w = 0.8$
$\sigma_w = 1.0$
$\sigma_w = 1.5$
$\sigma_w = 3.0$
orthogonal W
0
20
40
60
80
100
time gap k
LSTM cell state
$\|\partial(\mathbf{q}^T\mathbf{c}_T)/\partial\mathbf{c}_{T-k}\|$
10^{10}
10^{6}
10^{2}
10^{-2}
10^{-6}
10^{-10}
10^{-14}
10^{-18}
10^{-22}
$b_f = 0$
$b_f = 1$
$b_f = 3$
$b_f = 5$
0
20
40
60
80
100
time gap k